

The utilization of intraosseous BVN ablation to address vertebrogenic LBP has become a recognized safe, predictable, and durable surgical method for the management of chronic axial low back pain (LBP) identified using well-established clinical and magnetic resonance imaging findings, Modic type 1 and/or type 2 changes.

Intraosseous ablation of the BVN from the L3 through S1 vertebrae may be considered medically indicated for individuals with CLBP when all the following criteria are met:

Chronic low back pain (CLBP) of at least 6 months duration.

Failure to respond to at least 6 months of nonsurgical management.

[MRI]-demonstrated* MC1 or MC2 in at least 1 vertebral endplate at 1 or more levels from L3 to S1.

(*Endplate changes, inflammation, edema, disruption, and/or fissuring.)

Fibrovascular bone marrow changes (hypointense signal for Modic type 1).

Fatty bone marrow changes (hyperintense signal for Modic type 2)." (p. 7)

BVNA is NOT indicated in the following:

Patients with severe cardiac or pulmonary compromise.

Presence of implanted pulse generator(s) (e.g., pacemaker and defibrillator)/electronic implants except for circumstances where a specific patient safety precaution may be implemented.

Co-existence of other obvious radiographic etiology for patient's axial CLBP requiring a medically necessary surgical intervention.

Active or chronic infection, systemic or local.

Patients who are pregnant.

Skeletally immature patients (generally age <18 years).

Current or post-trauma, tumor, infection, or poor bone quality compromising vertebral pedicle/body.

Cauda equina syndrome defined as neural compression causing neurogenic bowel or bladder dysfunction.

Radiographic confirmation of gross spinal instability including angular or translatory instability (grade 2 or greater spondylolisthesis) at index level(s).

Morbid obesity precluding satisfactory procedural imaging.

Targeted ablation zone is <10 mm away from a sensitive structure not intended for ablation.

Situation where unintended tissue damage may result based on the clinical assessment by the physician.

The Centers for Medicare and Medicaid Services

LCD: L39420 Thermal Destruction of the Intraosseous Basivertebral Nerve (BVN) for Vertebrogenic Lower Back Pain

Approved: March 5, 2023

Covered Indications

Thermal destruction of the intraosseous BVN will be considered medically reasonable and necessary for the treatment of cLBP in patients who meet ALL the following criteria:

1. Chronic lumbar back pain of ≥ 6 months duration that causes functional deficit measured on a pain or disability scale*, AND
2. Documented failure to respond to ≥ 6 months of non-surgical management**, AND